

Nelson 50 km Fire Simulation Report

Summer 2025 weather-conditioned Cell2Fire scenario

Technical report | Generated 2026-07-22

Main result

This run produced a sparse cell-level burn-probability surface: the highest estimated cell probability was 1.4% (14 of 1,000 runs). That value is internally consistent with a six-hour, one-ignition-per-run screening experiment; it is not an annual probability or an operational fire forecast.

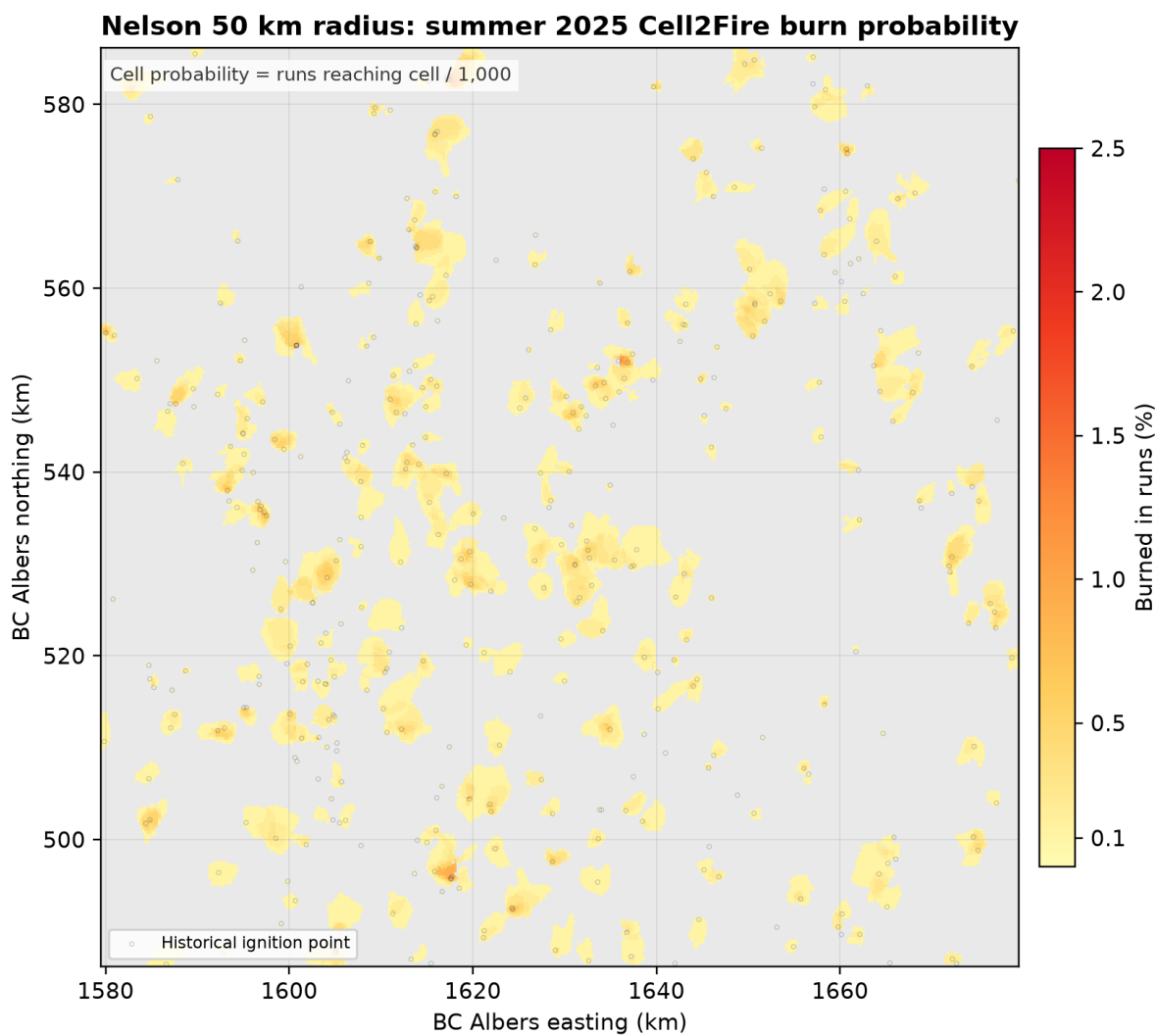


Figure 1. Static probability map from the latest completed run. The fixed color scale is shown in percent; empty grey cells are zero or no modeled burn. Historical ignition points are shown as small blue outlines.

Study area	Nelson, British Columbia; 50 km radius
Weather window	June 1-August 31, 2025; 184 accepted station-date scenarios
Landscape	250 m current fuel, elevation, slope, and aspect layers; EPSG:3005

Technical summary: the map is a conditional six-hour spread result

The latest scenario uses observed 2025 BC Wildfire Service weather records from the Smallwood and Slocan stations, sampled historical ignition locations from 2000-2023, and a current 250 m landscape. The pipeline completed all 1,000 requested Cell2Fire runs and aggregated final burned cells into a probability raster.

1,000 Completed runs <i>requested: 1,000</i>	33.69 cells Mean final footprint <i>2.11 km2 at 250 m</i>	1.4% Maximum cell probability <i>14 of 1,000 runs</i>	20,140 Cells reached at least once <i>12.6% of grid</i>
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What the result establishes

- The summer date filter worked as configured: 184 scenarios were retained, representing 92 dates at each of two stations.
- The summer-conditioned runs produced larger average footprints than the earlier all-season 2025 run: 33.69 versus 22.79 cells, when compared on the same 1,000-run design.
- The highest probability is low because the denominator is 1,000 single-fire trials and each trial lasts six one-hour periods. One hit is 0.1 percentage points; 14 hits is 1.4%.
- The probability surface should be read as conditional spread likelihood under this sampling design, not as the chance of at least one fire somewhere in the Nelson area during a year or summer.

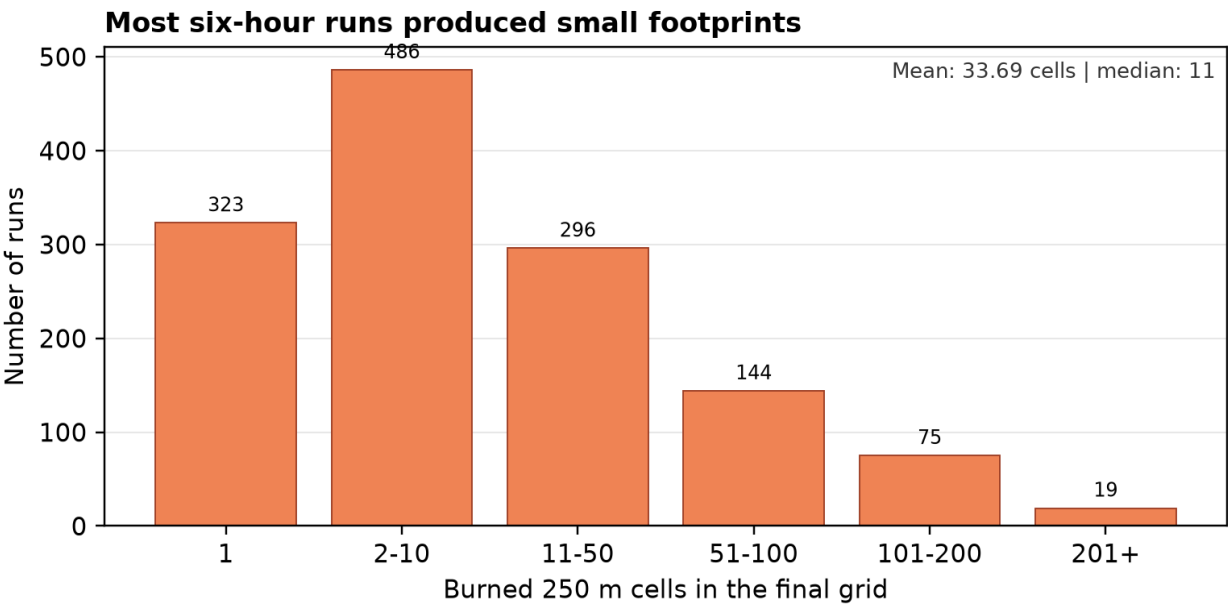


Figure 2. Distribution of final burned-cell counts across the 1,000 independent six-hour trials. A burned cell is a 250 m grid cell marked burned in the final Cell2Fire grid.

Scope, data, and metric definitions

The analysis covers a 50 km radius around Nelson, British Columbia. The raster processing envelope is 400 by 400 cells at 250 m resolution, or 160,000 cells before any circular study-area masking. The configured analysis CRS is BC Albers (EPSG:3005).

Item	Definition used in this report
Study area	Nelson centre 49.4928 N, -117.2948 W; configured radius 50,000 m.
Grid	400 x 400 cells; 250 m cell size; 160,000 raster cells.
Weather cohort	June 1-August 31, 2025; six hourly records per scenario, 12:00-17:00 local time.
Ignition cohort	408 historical BCWS point records; source fire years 2000-2023; 363 unique ignition cells appeared in the sampled runs.
Burn probability	For each cell: number of completed runs with a burned final-grid value of 1 divided by the number of completed runs.
Burned area	Number of burned cells multiplied by 0.0625 km2 per 250 m cell.

Weather scenarios and source coverage

Station	Accepted scenarios	Dates	FWI min	FWI mean	FWI max
SLOCAN	92	2025-06-01 to 2025-08-31	0.0	15.8	35.7
SMALLWOOD	92	2025-06-01 to 2025-08-31	0.0	20.3	41.1
Combined	184	2025-06-01 to 2025-08-31	0.0	18.1	41.1

The weather source began as BCWS DataMart records in long-term/nelson-50km/data/bcws-weather/2025_nelson_stations.csv. The preparation workflow retained complete six-hour noon-to-17:00 sequences and complete noon FWI-system values. The latest run retained only the summer date range; it did not create a continuous chronological summer weather record.

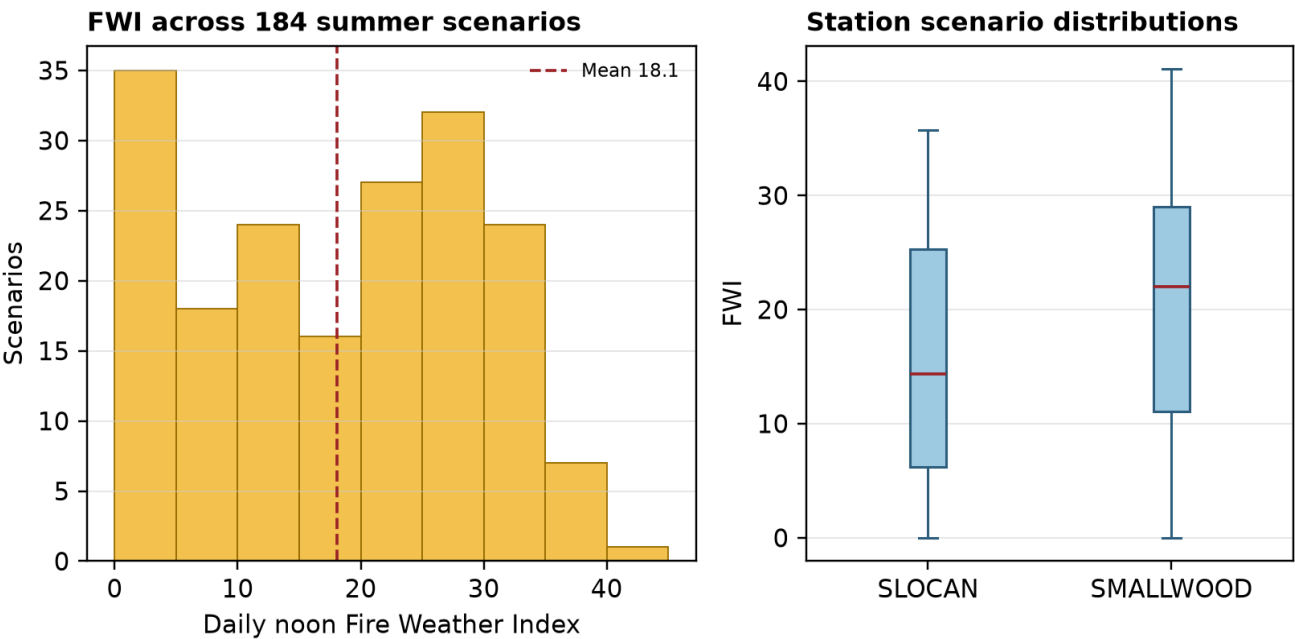


Figure 3. Distribution of daily noon FWI values used as weather scenarios. The FWI values are retained for provenance and reporting; the principal spread inputs in this configuration are fuel type, wind speed, wind direction, FFMC, and BUI.

Methodology and experimental design

The pipeline prepares Cell2Fire inputs from aligned rasters and source tables, samples weather and ignition records with a fixed random seed, executes independent runs, then aggregates the final burned-cell grids. Each run has one ignition and one six-row weather file.

Aligned inputs	Date-filter summer scenarios	Sample one weather + one ignition	Run Cell2Fire for 6 periods	Aggregate burn counts
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Setting	Configured value	Role in the run
Runs	1,000	Monte Carlo sample size and probability denominator.
Random seed	20250722	Makes weather and ignition sampling reproducible.
Weather records	6 rows per scenario	Consumed sequentially as six one-hour weather periods.
Fire period length	1.0 hour	One Cell2Fire fire period; six periods are requested.
ROS-CV	0.0	Deterministic rate-of-spread setting for this screening run.
Workers	4	Parallel execution only; does not alter model inputs.
Fuel / terrain	250 m; EPSG:3005	Fuel class, elevation, slope, and aspect are aligned to the same grid.

Sampling coverage and denominator

The run design contains 408 historical ignition records and 184 accepted weather scenarios, or 75,072 possible ignition-weather pairs. The pipeline sampled 1,000 pairs with replacement, so the realized manifest contains 363 unique ignition cells and 183 unique weather scenarios. The aggregate raster divides each cell's burn count by the 1,000 completed runs; therefore the smallest non-zero estimate is $1/1,000 = 0.1$ percentage points.

Ignition points that landed on non-burnable cells were snapped to the nearest burnable cell before being written to Cell2Fire Ignitions.csv. Historical fire number, year, and cause are provenance fields; the run did not apply cause-specific ignition weights.

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python3 fire_sim_pipeline.py --config examples/nelson-50km-2025-summer.json --stage all
```

The command above is the reproducible full-pipeline invocation from the repository root. The Cell2Fire runner is called once per manifest row with --ignitions, --weather rows, --max-fire-periods 6, --finalGrid, and a run-specific seed derived from 20250722.

Spatial result: burn activity is concentrated and low in absolute probability

The probability raster contains 20,140 cells with at least one burn hit and 139,860 cells with zero hits. The maximum cell probability is 1.4%, equal to 14 hits in the completed ensemble. The fixed color scale on the map is intentionally absolute: a visually warm cell is not automatically a 5% or 10% probability cell.

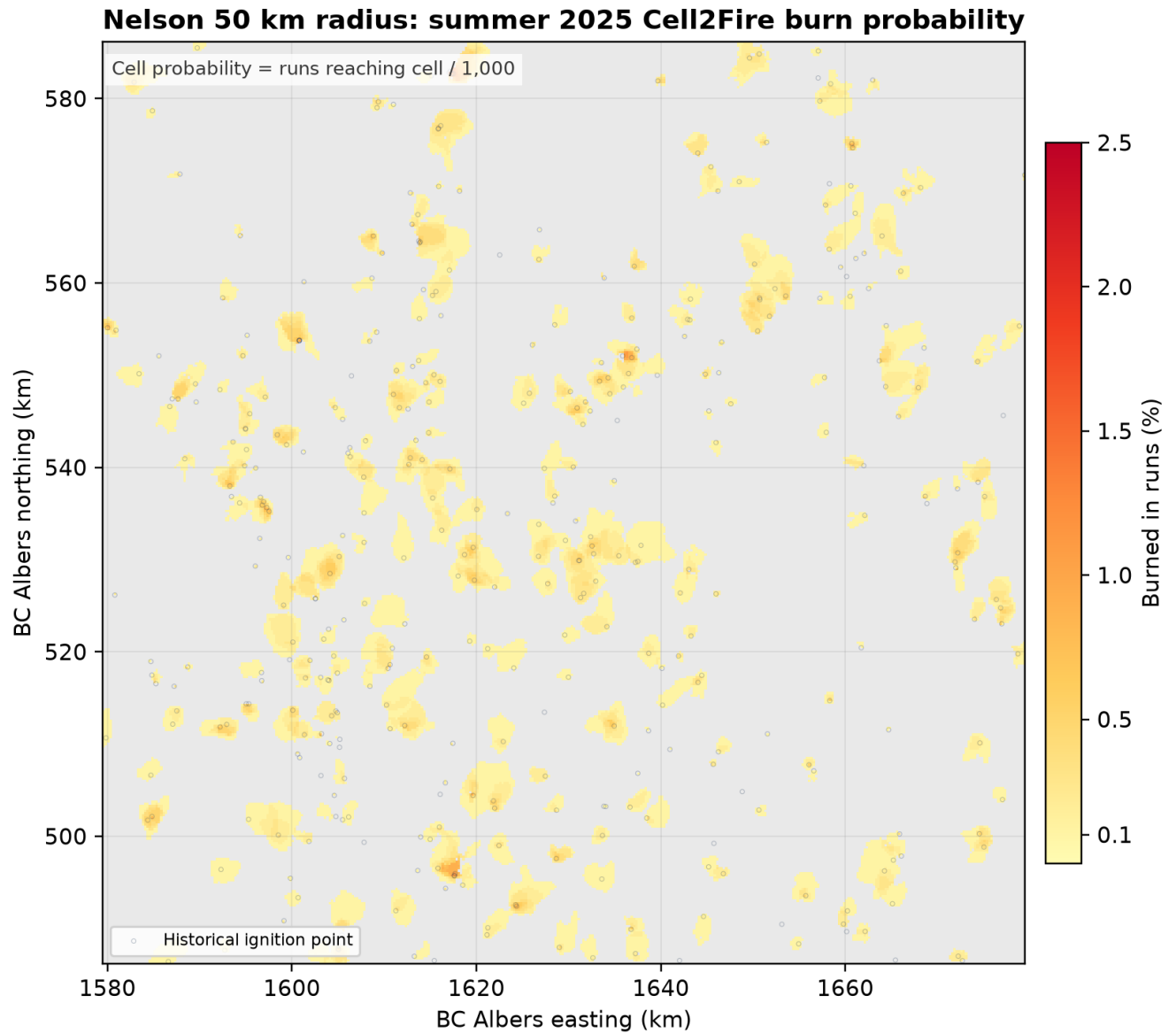
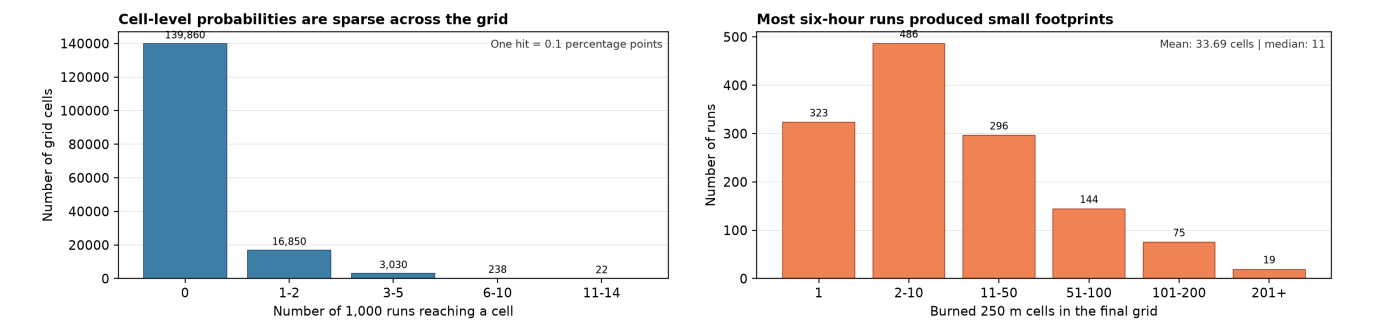


Figure 4. Burn-probability surface from the 250 m final-grid aggregation. The map is conditional on the selected ignition and weather sampling rules and six-hour simulation horizon. It should not be interpreted as an annual hazard map.

Spatial diagnostic	Value	Interpretation
Grid cells	160,000	Full 400 x 400 processing grid.
Non-zero cells	20,140 (12.6%)	Reached in at least one simulated run.
Maximum probability	1.4% (14/1000)	Highest empirical burn fraction for one cell.
Mean probability across grid	0.021%	Equivalent to the mean final footprint divided by the full grid.
Mean final area	2.11 km2	33.69 cells x 0.0625 km2 per cell.

Results and comparison: summer weather increased spread, but not the probability scale

The summer-only scenario produced a larger mean footprint than the earlier 2025 all-season scenario, which is directionally consistent with summer conditions. The cell-level probability scale remains low because both scenarios use one ignition and a six-hour window per run.



Figures 5-6. Left: most cells have zero or only one to two burn hits. Right: most runs have small final footprints, with a long tail of larger runs.

Metric	Summer 2025	All-season 2025	Change / reading
Completed runs	1,000	1,000	Same denominator.
Mean burned cells	33.69	22.79	+47.8%
Maximum probability	1.4%	1.3%	+0.1 percentage points
Cells with non-zero probability	20,140	15,191	Summer reached more cells in this sample.

Calculation spot-checks

Check	Result
Completed versus requested runs	1,000 / 1,000; all requested runs completed.
Maximum probability	max burn count 14 / 1,000 = 0.014 = 1.4%.
Mean footprint conversion	33.69 cells x 0.0625 km2 = 2.11 km2.
Weather filter	184 scenarios x 6 hourly rows = 1,104 unique prepared weather rows.
Run manifest	1,000 rows; 183 unique weather files and 363 unique ignition cells realized.

Limitations, uncertainty, and robustness checks

Overall assessment: share with caveats. The run is reproducible and internally consistent as a planning-screening experiment, but it is not calibrated or designed to estimate annual wildfire risk.

Issue	Why it matters
Short simulation horizon	Each fire is simulated for six one-hour periods. A longer-lived fire can travel farther and affect more cells, so these probabilities are conditional on a short spread window.
One ignition per run	The ensemble does not simulate multiple ignitions during a summer or year. It therefore cannot answer the cumulative probability that a cell burns at least once during a season.
Sampling noise	With 1,000 runs, estimates move in 0.1 percentage-point steps. Most non-zero cells have only one to two hits, so local rankings are unstable without a larger ensemble.
Weather chronology	Each run samples one six-hour station-date scenario. The weather library is not a continuous chronological summer sequence, and scenarios are sampled uniformly rather than weighted by fire occurrence.
Ignition model	Historical ignition points are sampled uniformly from the available point records. Fire-cause, year, and spatial intensity differences are not used as weights in this run.
Landscape and calibration	The landscape is a current 250 m fuel/topography representation. The run has not been calibrated against observed historical fire perimeters or validated with an independent holdout.
Operational use	Outputs are for planning and research only. They are not a live-fire forecast, evacuation product, or operational decision surface.

Robustness evidence

The summer filter is supported by the manifest (184 accepted scenarios from June 1-August 31, 2025), and the output denominator is reconciled to all 1,000 completed runs. Compared with the prior all-season 2025 run, summer increased mean final footprint from 22.79 to 33.69 cells and maximum probability from 1.3% to 1.4%. These are useful directional checks, but they do not replace convergence testing, calibration, or a longer-duration experiment.

Recommended next steps

- Choose the target metric explicitly: conditional spread probability per ignition, annual cumulative probability, or probability of exceeding a burned-area threshold.
- For conditional spread, generate continuous 24-72 hour weather sequences and increase the ensemble to at least 10,000 runs, with stratified or exhaustive coverage of ignition cells and weather scenarios.
- For annual risk, model the number of ignitions per synthetic year from historical BCWS rates and simulate multiple starts per year across a full fire season.
- Replay a calibration set of observed fire perimeters, then report sensitivity to fuel mapping, wind, ignition weighting, duration, and run count.
- Keep the absolute probability scale in the map, but add a separate quantile or log-stretch view only for visual exploration; do not relabel display classes as higher probabilities.

Further questions

The key unresolved question is whether this project should represent spread conditional on a fire start or cumulative annual risk. That decision determines the ignition process, weather chronology, simulation duration, and probability denominator for the next run.

Appendix: reproducibility and source inventory

The following files are the primary audit trail for this report. Paths are relative to the repository root.

Artifact	Purpose
examples/nelson-50km-2025-summer.json	Complete scenario configuration, including summer date bounds and run settings.
fire_sim_pipeline.py	Config-driven preparation, Cell2Fire execution, aggregation, and web-map build.
Cell2Fire/data/nelson-50km-2025-summer-reusable/run_manifest.csv	The 1,000 sampled weather-ignition pairs.
Cell2Fire/data/nelson-50km-2025-summer-reusable/weather_manifest.csv	The 184 accepted station-date scenarios.
Cell2Fire/data/nelson-50km-2025-summer-reusable/Weathers/	Six hourly records per accepted scenario.
runs/nelson-50km-2025-summer/outputs/burn_probability.tif	Cell-level empirical probability raster.
runs/nelson-50km-2025-summer/outputs/burn_count.tif	Cell-level burn-hit count raster.
runs/nelson-50km-2025-summer/outputs/run_summary.csv	Burned-cell count for every completed run.
runs/nelson-50km-2025-summer/web-map/index.html	Interactive local Leaflet map with probability, fuel, ignition, and basemap layers.
runs/nelson-50km-2025-summer/report-assets/weather_data_used.csv	6,000 weather rows joined to the 1,000 run manifest rows for this report.

Key output checksums

File	SHA-256 prefix
runs/nelson-50km-2025-summer/outputs/burn_probability.tif	c744d55d36280c81
runs/nelson-50km-2025-summer/outputs/burn_count.tif	0ab62ffb252780fc
runs/nelson-50km-2025-summer/outputs/metadata.json	bc08620288001020
Cell2Fire/data/nelson-50km-2025-summer-reusable/run_manifest.csv	2bb67cde21adb245
Cell2Fire/data/nelson-50km-2025-summer-reusable/weather_manifest.csv	5123586a53b743b4

Weather field note

The exported weather CSV retains the original six-hour rows and the run-level ignition provenance. Temperature, relative humidity, and precipitation are retained for traceability; in this configuration, the main spread-driving fields are wind speed, wind direction, FFMC, and BUI. Daily noon FWI-system values are repeated across that day's six rows by the prepared Cell2Fire weather files.

Report generation note

This PDF was generated from the completed run artifacts on 2026-07-22. It is a static report snapshot; the interactive web map remains the best surface for toggling probability, fuel, ignition, and basemap layers.